

Motion-Enabled Tomography via Gaussian Mixture Models*

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Abstract. Recovering physical properties of objects in motion is a core task across scientific and industrial applications. When the relative motion between the object and the sensing apparatus provides sufficient angular coverage, Computerized Tomography offers a powerful means of reconstruction. For such scenarios, we propose a parametric spatiotemporal model applied to Gaussian Mixture Models (GMM), in which each constituent Gaussian is parameterized by its own angular velocity, projectile motion, and geometry. GMM are a suitable means of reconstruction because they (i) admit accurate approximations in object space and (ii) have a closed form expression under the ray transform; enabling efficient forward predictions and exact gradient computations in data space. By decoupling the reconstruction problem into two sub-inverse problems, we characterize solutions as minimizers of task-specific objective functions that are derived and solved by utilizing the properties of (ii). The resulting algorithm we provide is applicable to objects in Euclidean space of arbitrary dimension. We validate the method on a simulated 2D problem, achieving accurate reconstruction of a 5-Gaussian GMM with intersecting trajectories. This also provides a foundation for further experimentation in settings with noisy data, 3D objects, and non-rigid body dynamics.

Key words. Computerized tomography, Gaussian mixture model, Radon transform, object reconstruction.

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1. Introduction. Tomography refers to imaging technologies that enable visualization of the internal structure of objects in a non-destructive manner. The basic idea is to repeatedly probe the object with waves or particles from different directions, measure how they are altered after passing through or interacting with the object, and use these measurements to reconstruct a cross-sectional or three-dimensional image of its interior. In many applications, it is however desirable to recover a model of the object. A “model” refers here to a representation of the object that goes beyond the traditional image format, which typically consists of an array of pixel or voxel values. Next, some applications involve tomographic imaging of objects that move during the data acquisition. If the motion is unknown, then it needs to be estimated.

An example of tomographic imaging that involves both model building and motion estimation is cryo-EM single-particle analysis, where tomographic imaging principles are used to reconstruct 3D molecular structures from a series of 2D transmission electron microscope images. Assuming homogeneous particles, the reconstruction problem amounts to recovering the atomic structure of a protein (model) from a collection of 2D electron microscopy images. Each 2D image represents tomographic projection data of the protein in an unknown 3D orientation, so the process can be understood as tomographic imaging of a protein that undergoes unknown rotations (motion) during data acquisition. Other examples of such imaging problems arise in industrial tomographic imaging where it is crucial to acquire data quickly while keeping the instrumentation as simple and cost-efficient as possible. One way to reduce costs is to use a fixed source–detector setup and allow the object to move through the field of view. Ideally, this motion is controlled; however,

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in low-cost scenarios it may be uncontrolled, e.g., a freely falling object undergoing rotation. This calls for joint motion estimation and reconstruction. Additionally, one can also include model building to represent the objects being imaged in a more appropriate way. The use case we have as the underlying motivation for the work in this paper comes from industrial tomography. It deals with computerized tomography (CT) imaging for subsurface characterization. The instrumental setup leads to tomographic reconstruction of particles that undergo unknown motions. We are primarily interested in recovering the morphological shape of the particles, so they can be modeled by a Gaussian mixture model (GMM). Sections 1.1–1.2 provide further details and Section 1.3 outlines the challenges associated with this use case.

The common approach for tomographic reconstruction with a down-stream post-processing step, like model building, is sequential: first reconstruct an image of the object and its internal structure, then use this image as input to a separate post-processing step that builds the desired model. Alternatively, one can integrate the reconstruction and post-processing steps, i.e., jointly perform reconstruction and model building. One can adopt a similar sequential approach for reconstruction with an up-stream pre-processing step, like motion estimation. Alternatively, just as with post-processing, one can integrate the motion estimation and reconstruction steps, i.e., perform motion estimation jointly with image reconstruction.

The approach considered in this paper is to jointly perform the following three tasks: motion estimation, tomographic reconstruction, and model building. Such a joint approach may appear more complex than a sequential one, but it also opens up for using more efficient regularization methods. This is needed as tomographic reconstruction of an object that undergoes an unknown motion is a severely ill-posed inverse problem. A key advantage that comes with a joint formulation that also includes model building is that the latter can act as a form of regularization for the motion estimation and reconstruction. However, this benefit depends on adopting a carefully chosen representation. In our framework, the object is modeled using a Gaussian mixture model (GMM), where it is expressed as a linear combination of Gaussian components. This is a natural model in many uses cases, e.g., for representing proteins where atoms are represented by Gaussians [39] or in a multi-scale setting where Gaussians represent residues [4]. GMM have therefore found widespread use in the above mentioned cryo-EM single-particle analysis [11, 36, 12] and electron tomography [5, 53]. Another important application area is in computer vision and robotics, where GMM-based representations play a central role in many successful deep learning approaches to 3D scene reconstruction [20, 43, 24]. A notable example is the usage of neural radiance fields (NeRFs), which recover 3D scene representations in the form of GMM from 2D images captured from multiple viewpoints [47, 22, 48, 18, 7, 50]. On a final note, we also mention that the joint approach taken in this paper is part of an active research area within inverse problems on developing theory and algorithms methods for jointly solving an ill-posed inverse problem (reconstruction) with a downstream post-processing task [30, 1]. Similarly, it also aligns with the current drive towards multimodal learning, which is a type of deep learning that integrates and jointly processes multiple different data types (modalities), like text, audio, images, or video [29, 51, 52].

1.1. Subsurface characterization. One of the main tasks in the reservoir characterization process is to determine petrophysical properties of the excavated rocks (porosity, saturation, permeability) early in the process. This enables one to characterize the reserves as well as their productivity and deliverability. Although logs generally provide a good estimate of porosity and saturation along the well, permeability is harder to assess especially in carbonates. Drill cuttings can provide lithological and other additional information on the petrophysical properties of the reservoir [28].

Lithology interpretation refers to the process to identify and interpret lithology of the geological layers from surface and subsurface drilling measurements. This plays a foundational role in subsurface exploration and drilling operations, whether for geothermal energy or conventional oil and gas resources. It involves the identification and characterization of rock types and their properties, which directly influence drilling strategy, reservoir evaluation, and resource extraction efficiency. Accurate lithological interpretation enables engineers and geoscientists to optimize well placement, anticipate drilling hazards, and assess the viability of subsurface formations.

Lithology interpretation is particularly critical in geothermal drilling for locating heat-bearing formations and understanding the thermal and hydraulic behavior of the subsurface. It helps to identify permeable zones, estimate temperature gradients, and guide well design to maximize energy recovery. In conventional oil and gas drilling, lithology interpretation informs reservoir characterization, petrophysical analysis, and structural mapping. It supports the identification of productive zones, seal integrity, and hydrocarbon potential, while also aiding in the mitigation of drilling risks such as overpressured formations or unstable shales [26, 35, 44].

One of the most widely used methods for lithology identification is visual analysis of drilled cuttings. Drilled cuttings are often angular, fractured, and irregular due to the mechanical action of drilling. Geologists characterize them through a combination of visual inspection, physical testing, and analytical techniques. Cuttings examined under the microscopes are characterized by their grain size and shape, color and texture, mineral composition, cementation and porosity. Roundness, sphericity, aspect ratio, convexity, and solidity are among the shape descriptors.

While cost-effective and accessible, the above approach is labor-intensive due to the frequent manual collection and preparation of cuttings, fragmented nature and collected volume of cuttings, potential contamination, and the need for expert interpretation. Geologists must visually inspect samples, often under microscopes, and integrate findings with log data and regional geological models. Despite advancements in automated mineralogy and machine learning, manual interpretation remains essential, especially in complex geological settings [49, 32, 9].

1.2. Tomographic imaging in subsurface characterization. Motivated by reducing the aforementioned labor intensive effort in collection, preparation, and interpretation process, we conceptualize a system for online tomographic imaging of the cuttings as they are being discharged into a collection pit.

CT imaging has been successfully used to get structural information of materials in a non-destructive fashion. Its usage to image micro-cores is almost a standard procedure in the energy industry and is referred to as “digital rock” [38, 31, 2, 3, 6, 40]. In digital rock, microscopic imaging obtained from CT scans are combined with numerical simulations to analyze the physical properties of rocks and helps to infer about petrophysical properties of reservoir. This is limited to small numbers of core samples. If upscaled to the whole well with high throughput, it could help determination of geomechanical properties of the subsurface [14] that could aid well construction and complement well integrity measurements. A similar approach was also investigated for cement-based materials [13].

Incident energy dependency of the attenuation of the object can be used as a spectroscopic method [41]. When combined with the structural information, this could enable mineral characterization [46]. X-ray attenuation depends on the density of the material and its atomic composition, which vary significantly between different rock types. For example, carbonates (e.g., limestone, dolomite) have moderate attenuation due to calcium and magnesium content, shales have higher attenuation if rich in iron or organic mat-

142 ter. Sandstones have lower attenuation unless cemented with heavy minerals, and igneous
143 rocks have high attenuation due to dense minerals like feldspar, pyroxene, and olivine [37].

144 **1.3. Challenges.** There are hardware and software challenges associated with devel-
145 oping a tomographic imaging system that can allude to the structural and attenuation
146 properties of cuttings.

147 One is to have a cost effective acquisition system, e.g., by considering a gantry with
148 fixed source and detector elements. Performing tomographic imaging in this setting relies
149 on letting the objects being imaged to rotate as they move through the gantry. Another
150 is to design an associated reconstruction algorithm that is suited to deal with the high
151 throughput of the system while delivering the highest image quality with least computa-
152 tional load for real-time applications.

153 Tomographic imaging of objects in motion has been a subject of investigation with
154 fixed acquisition geometries and multiple sources [33, 19]. The inherent ill-posed nature
155 of tomography problems makes it impossible to completely recover static objects with a
156 fixed single-source acquisition geometry. However, we assume that discharged objects un-
157 dergo parabolic and rotational motion while they traverse the field of view of the imaging
158 system. This assumption transforms the limited-angle tomography into a finite angle to-
159 mography with the additional complexity of recovering dynamic properties of the object in
160 conjunction to its reconstruction. In this study we will focus on the dynamic tomographic
161 reconstruction problem for a given detector geometry and leave the optimal experimental
162 design problem [16] of using a minimal number of detectors for a future study.

163 The associated software challenge is to investigate whether it is possible to image dis-
164 charged cuttings as they go under parabolic motion. We consider approximation cuttings
165 as GMM. A motivation for this representation is that it has the capacity to approximate
166 general objects up to an arbitrary degree of accuracy [55]. Another motivation is that the
167 ray transform of this representation is a lower dimensional GMM which can be analytically
168 computed. For the sake of simplicity we approximate each cutting by a single Gaussian.
169 Characterizing the shape of a drilled cutting as a Gaussian is not typically done in geology,
170 but it provides convenience from a mathematical modeling or image analysis perspective.
171 As a consequence, discharge of multiple cuttings constitutes a dynamic GMM. This is a
172 reasonable first step in our feasibility study before approximation of individual cuttings
173 with GMM. In this form, describing outline, cross-section, curvature or thickness by fitting
174 a 3D Gaussian blob can approximate the volume or morphology of a cutting particle.

175 **2. The mathematical setting.** Consider performing tomography on an object that
176 undergoes motion. The starting point for the mathematical formalization is to represent
177 the object with a real valued function $f: \mathbb{R}^d \times [0, t_{\max}] \rightarrow \mathbb{R}$. Data is similarly given as
178 (noisy) samples of a function $g: \mathbb{M} \times [0, t_{\max}] \rightarrow \mathbb{R}$ where $\mathbb{M}$ is some manifold of lines in
179 $\mathbb{R}^d$ and

$$180 \quad (2.1) \quad g(\ell, t) := \mathcal{X}(f(\cdot, t))(\ell) := \int_{\mathbf{x} \in \ell} f(\mathbf{x}, t) d\mathbf{x} \quad \text{for } \ell \in \mathbb{M},$$

181 with $\mathcal{X}$ denoting the ray transform ($d\mathbf{x}$ is the 1-dimensional Lebesgue measure on the line
182 ℓ).

183 **Definition 2.1 (Spatiotemporal imaging).** Recover $t \mapsto f(\cdot, t)$ on $\mathbb{R}^d$ from time-series
184 $t \mapsto g(\cdot, t)$ on $\mathbb{M}$ where $g: \mathbb{M} \times [0, t_{\max}] \rightarrow \mathbb{R}$ is given as in (2.1).

185 The data collected by the experimental source/detector setup corresponds to sampling
186 on the manifold $\mathbb{M}$ of lines in $\mathbb{R}^d$. Our experimental setup has a *fixed* point-like source
187 (source position) with fixed detectors that are represented as a planar rectangular region

contained in a $(d-1)$ -dimensional plane (detector plane). We assume the source is centered w.r.t. the detector, i.e., projecting the source position onto the detector plane coincides with the midpoint of the rectangular region that makes up the detector.

The above inverse problem is clearly under-determined as it involves recovering a real-valued function on $\mathbb{R}^d$ from data given as a real-valued function on $\mathbb{M}$, which is the family of lines intersecting a $(d-1)$ -dimensional rectangular region (detector) and that go through a point (source position). We therefore need to make use of further assumptions.

In the application we have in mind, we can assume that the object consists of N rigid isolated particles that move independently of each other. Hence, the temporal variation of the object corresponds to N independent motion trajectories, one for each particle. Since each particle undergoes a time dependent rigid body motion, we can re-interpret this as viewing each particle from directions that vary with time, i.e., as tomographic imaging with an unknown acquisition geometry.

To formalize the above, the function $f: \mathbb{R}^d \rightarrow \mathbb{R}$ representing the object can be written as

$$(2.2) \quad f(\mathbf{x}, t) = \sum_{n=1}^N f_n(\mathbf{x}, t) \quad \text{for } (\mathbf{x}, t) \in \mathbb{R}^d \times [0, t_{\max}]$$

where $f_n: \mathbb{R}^d \rightarrow \mathbb{R}$ represents the n -th particle. The assumption that each particle undergoes a time-dependent rigid body motion is then encoded as

$$(2.3) \quad f_n(\mathbf{x}, t) := (\mathbf{M}_n(t) \cdot \rho_n)(\mathbf{x}),$$

where $\rho_n: \mathbb{R}^d \rightarrow \mathbb{R}$ represents the particle structure and the particle-specific rigid body motion is represented by the action $(\mathbf{M}, \rho) \mapsto \mathbf{M} \cdot \rho$ of $\mathbf{M} \in \text{SE}(d)$ on the set of $\mathbb{R}$ -valued functions on $\mathbb{R}^d$. The latter is given as

$$(2.4) \quad (\mathbf{M} \cdot \rho)(\mathbf{x}) := \rho(\mathbf{M} \cdot \mathbf{x}) = \rho(\mathbf{R}^{-1}(\mathbf{x} - \boldsymbol{\tau})) \quad \text{for } \mathbf{M} = (\mathbf{R}, \boldsymbol{\tau}) \in \text{SE}(d) = \text{SO}(d) \ltimes \mathbb{R}^d.$$

Inserting (2.3) into (2.2) yields the following spatiotemporal model for the object:

$$(2.5) \quad f(\mathbf{x}, t) = \sum_{n=1}^N (\mathbf{M}_n(t) \cdot \rho_n)(\mathbf{x}) = \sum_{n=1}^N \rho_n(\mathbf{R}_n(t)^{-1}(\mathbf{x} - \boldsymbol{\tau}_n(t))),$$

where each curve $t \mapsto \mathbf{M}_n(t) = (\mathbf{R}_n(t), \boldsymbol{\tau}_n(t)) \in \text{SE}(d)$ models the motion of the n -th particle. Inserting the motion model (2.5) into (2.1) yields the following model for the data:

$$(2.6) \quad g(\ell, t) := \mathcal{X}\left(\sum_{n=1}^N \mathbf{M}_n(t) \cdot \rho_n\right)(\ell) = \sum_{n=1}^N \mathcal{X}(\mathbf{M}_n(t) \cdot \rho_n)(\ell),$$

where the last equality follows from the linearity of the ray transform.

As a generative model for the morphology component of the particles, we assume they are all dilations of a *single* template ρ_0 , i.e. morphology of particles are in the $\text{GL}(d)$ orbit of ρ_0 :

$$(2.7) \quad \text{GL}(d)\rho_0 := \{\mathbf{U}_n^{-1} \cdot \rho_0 \mid \mathbf{U}_n \in \text{GL}(d)\} = \{\mathbf{x} \mapsto \rho_0(\mathbf{U}_n \mathbf{x}) \mid \mathbf{U}_n \in \text{GL}(d)\},$$

where $\text{GL}(d)$ is the general linear group on $\mathbb{R}^d$.

Remark 1. Note that $\mathbf{U}^{-1} \cdot \rho_0: \mathbb{R}^d \rightarrow \mathbb{R}$ depends only on the symmetric non-negative definite matrix $\mathbf{U}^\top \mathbf{U}$, so we can without loss of generality assume that $\mathbf{U} \in \mathbb{R}^{d \times d}$ is upper triangular with non-negative diagonal entries, $\mathbf{U} \in \mathbb{R}^{d_U} \subset \mathbb{R}^{d \times d}$, where $d_U = d(d+1)/2$

For example, if morphology component is given by an isotropic Gaussian $\rho_0(\mathbf{x}) = e^{-\mathbf{x}^\top \mathbf{x}}$, then

$$(2.8) \quad \rho_n(\mathbf{x}) := \alpha_n(\mathbf{U}_n^{-1} \cdot \rho_0)(\mathbf{x}) = \alpha_n \rho_0(\mathbf{U}_n \mathbf{x}).$$

Next, combination of (2.2), (2.3), and (2.8) constitutes a Gaussian mixture model (GMM) model for $\mathbf{x} \mapsto f(\mathbf{x}, t)$. A motivation for this representation is that it has the capacity to approximate general objects up to an arbitrary degree of accuracy [55]. Another motivation is that this representation is mapped to a lower dimensional GMM by the ray transform that can be analytically computed as will be shown in Proposition 5.2.

Substituting (2.8) into (2.5) and (2.6) yields the following generative model for the object and data, respectively,

$$(2.9) \quad f(\mathbf{x}, t) = \sum_{n=1}^N \alpha_n (\mathbf{M}_n(t) \cdot \mathbf{U}_n^{-1} \cdot \rho_0)(\mathbf{x}) \quad \text{for } \mathbf{x} \in \mathbb{R}^d \text{ and } t \in [0, t_{\max}].$$

and

$$(2.10) \quad g(\ell, t) := \mathcal{X} \left(\sum_{n=1}^N \alpha_n \mathbf{M}_n(t) \cdot \mathbf{U}_n^{-1} \cdot \rho_0 \right) (\ell) = \sum_{n=1}^N \alpha_n \mathcal{X} \left(\mathbf{M}_n(t) \cdot \mathbf{U}_n^{-1} \cdot \rho_0 \right) (\ell) \quad \text{for } \ell \in \mathbb{M} \text{ and } t \in [0, t_{\max}].$$

The inverse problem is now to determine N dilations $\mathbf{U}_n \in \mathbb{R}^{d_0}$ and N curves $t \mapsto \mathbf{M}_n(t)$ from a single time series $t \mapsto g(\cdot, t)$ of ray transform data on $\mathbb{M}$. Each $\mathbf{U}_n$ generates the structures for the particles through (2.8) and $t \mapsto \mathbf{M}_n(t)$ is a nuisance parameter representing the particle specific motions. We expect that this inverse problem is better posed than the original one stated after (2.1).

We next introduce computationally feasible parameterizations of the motion model (Section 3) and the ray transform (Section 5).

3. Parameterizing particle motions. An element $\mathbf{M} \in \text{SE}(d)$ can be parameterized as a combination of translation and rotation which in matrix form is expressible as

$$(3.1) \quad \mathbf{M} = \begin{bmatrix} \mathbf{R}_\Omega & \boldsymbol{\tau} \\ \mathbf{0}^\top & 1 \end{bmatrix}.$$

In the above, $\boldsymbol{\tau} \in \mathbb{R}^d$ is the translation, $\mathbf{0} \in \mathbb{R}^d$ is the zero vector, and $\mathbf{R}_\Omega \in \text{SO}(d)$ is parameterized by $\Omega \in \mathbb{R}^{d_R}$ with $d_R := d(d-1)/2$. One way to parameterize $\mathbf{R}_\Omega$ is as the product

$$(3.2) \quad \mathbf{R}_\Omega := \prod_{k=1}^{d_R} \mathbf{R}_k(\omega_k), \quad \text{for } \omega_k \in \mathbb{R}$$

where

$$(254) \quad \mathbf{R}_k(\omega_k) := \begin{bmatrix} \overset{d_1-1}{\boxed{\mathbf{I}_1}} & & & & & \\ & \cos \omega_k & & & -\sin \omega_k & \\ & & \boxed{\mathbf{I}_2} & & & \\ & \sin \omega_k & & \cos \omega_k & & \\ & & & & \boxed{\mathbf{I}_3} & \\ & & & & & \end{bmatrix} \begin{matrix} d_2-d_1-1 \\ d_R-d_2 \end{matrix}$$

for $k = (d_1-1)(d-d_1) + (d_2-d_1)$. Here $d_1 < d_2 \leq d$ is the row and column indices for specific elements in $\mathbf{R}_k(\omega_k)$ and

- $\mathbf{l}_1$ is the $(d_1 - 1) \times (d_1 - 1)$ identity matrix,
- $\mathbf{l}_2$ is the $(d_2 - d_1 - 1) \times (d_2 - d_1 - 1)$ identity matrix,
- $\mathbf{l}_3$ is the $(d_R - d_2) \times (d_R - d_2)$ identity matrix,

Besides the elements shown, all other elements of $\mathbf{R}_k(\omega_k)$ are zero.

One can now model the curve $\mathbf{M}_n(t) \in \text{SE}(d)$ by a time-dependent translation $\boldsymbol{\tau}_n(t) \in \mathbb{R}^d$ and rotation $\mathbf{R}_{\boldsymbol{\Omega}_n(t)} \in \text{SO}(d)$ given by

$$(3.3) \quad \boldsymbol{\tau}_n(t) = \boldsymbol{\mu}_n + t \mathbf{v}_n + \frac{t^2}{2} \mathbf{a}_n \quad \text{for some } \mathbf{v}_n, \mathbf{a}_n \in \mathbb{R}^d$$

$$(3.4) \quad \boldsymbol{\Omega}_n(t) := t \boldsymbol{\Theta}_n, \quad \text{for } \boldsymbol{\Theta}_n \in \mathbb{R}^{d_R}$$

where

$$\mathbf{R}_{\boldsymbol{\Omega}_n(t)} := \prod_{k=1}^{d_R} \begin{bmatrix} \mathbf{l}_1 & & & \\ & \cos(t\theta_{n,k}) & & -\sin(t\theta_{n,k}) \\ & & \mathbf{l}_2 & \\ & \sin(t\theta_{n,k}) & & \cos(t\theta_{n,k}) \\ & & & & \mathbf{l}_3 \end{bmatrix}$$

and with $\theta_{n,k}$ denoting the k -th component of $\boldsymbol{\Theta}_n$.

4. Parameterizing the domain of the forward operator. The parameterization in Section 3 allows us to rewrite the data model in (2.10). Let us employ the following notation:

- $(\alpha, \mathbf{U}) \in \mathbb{R} \times \mathbb{R}^{d_U}$ encodes the amplitude/attenuation and shape/morphology of the particle.
- $\boldsymbol{\Theta} \in \mathbb{R}^{d_R}$ encodes the rotational motion $\mathbf{R}_{t\boldsymbol{\Theta}} \in \mathbb{R}^{d_R}$ of the particle.
- $\boldsymbol{\eta} := (\boldsymbol{\mu}, \mathbf{v}, \mathbf{a}) \in \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d$ encodes the projectile motion curve $\mathbf{C}[\boldsymbol{\eta}]: [0, t_{\max}] \rightarrow \mathbb{R}^d$ of the particle defined as

$$\mathbf{C}[\boldsymbol{\eta}](t) := \boldsymbol{\mu} + t\mathbf{v} + \frac{t^2}{2} \mathbf{a}.$$

Here, $\boldsymbol{\Theta}$ is the angular velocities, $\boldsymbol{\mu}$ is the spatial location at time $t = 0$, $\mathbf{v}$ is the initial directional velocity, and $\mathbf{a}$ is the initial acceleration.

Let $\text{Aff}(d) = \text{GL}(d) \ltimes \mathbb{R}^d$ denote the affine group, and $\mathbf{A}[\mathbf{U}, \boldsymbol{\Theta}, \boldsymbol{\eta}](t) = (\mathbf{R}_{t\boldsymbol{\Theta}}\mathbf{U}^{-1}, \mathbf{C}[\boldsymbol{\eta}](t)) \in \text{Aff}(d)$ capture the morphology and motion of a particle. Denoting the n -th particle by

$$(4.1) \quad \mathbf{M}_n(t) \cdot \rho_n = \mathbf{A}[\mathbf{l}, \boldsymbol{\Theta}_n, \boldsymbol{\eta}_n](t) \cdot \rho_n = \alpha_n \mathbf{A}[\mathbf{U}_n, \boldsymbol{\Theta}_n, \boldsymbol{\eta}_n](t) \cdot \rho_0,$$

we see that the morphological components of the particles undergoing motion are in the affine group orbit of ρ_0 :

$$(4.2) \quad \text{Aff}(d)\rho_0 := \{\mathbf{A} \cdot \rho_0 \mid \mathbf{A} \in \text{Aff}(d)\} = \{\mathbf{x} \mapsto \rho_0(\mathbf{U}^{-1}(\mathbf{x} - \boldsymbol{\mu})) \mid \mathbf{A} = (\mathbf{U}, \boldsymbol{\mu}) \in \text{Aff}(d)\}.$$

Substituting (4.1) into (2.10) and using the above notation allows us to express the inverse problem stated after (2.10) as the task to recover

$$\Gamma := (\gamma_1, \dots, \gamma_N) \quad \text{with} \quad \gamma_n := ((\alpha_n, \mathbf{U}_n), \boldsymbol{\Theta}_n, \boldsymbol{\eta}_n) \in (\mathbb{R} \times \mathbb{R}^{d_U}) \times \mathbb{R}^{d_R} \times (\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d)$$

from data

$$(4.3) \quad g(\ell, t) = F[\Gamma](\ell, t) \quad \text{for } (\ell, t) \in \mathbb{M} \times [0, t_{\max}]$$

where $F[\Gamma]: \mathbb{M} \times [0, t_{\max}] \rightarrow \mathbb{R}$ (forward operator) is given as

$$(4.4) \quad F[\Gamma](\ell, t) := \sum_{n=1}^N \alpha_n \mathcal{X}(A[U_n, \Theta_n, \eta_n](t) \cdot \rho_0)(\ell).$$

To summarize, the original inverse problem stated right after (2.1) was to recover the time-dependent structure of the object from a time series of single projections, which is clearly severely ill-posed. To address the latter, we assume the object consists of particles that move independently from each other as expressed in (2.3). The corresponding inverse problem, which is stated after (2.10), is still under determined as it involves recovering particle structures (which are elements in an infinite dimensional space) and their motions (which is encoded as an element in an infinite dimensional space). For this, we use a generative model for the particle structures (2.8) and a parabolic motion model (3.3) with rotations (3.4). All of these are specified by finitely many parameters, so the corresponding inverse problem considering (4.3) is not necessarily under-determined. A drawback is that the corresponding forward operator, which was linear in (2.1), is now non-linear.

The above dealt with introducing coordinates on the domain of the forward operator, i.e., to parameterize time-dependent, $\mathbb{R}$ -valued functions on $\mathbb{R}^d$. The final step necessary for implementation is to introduce coordinates on the range of the forward operator, which is time dependent $\mathbb{R}$ -valued functions on $\mathbb{R}^d$ (time dependent sinograms). This is done in Section 5.

5. Parameterizing the range of the forward operator. The starting point is to introduce coordinates on the manifold of lines $\mathbb{M}$. A line $\ell \in \mathbb{M}$ can be parameterized by a source-detector pair of points $(\mathbf{s}, \mathbf{r}) \in \mathbb{R}^d \times \mathbb{R}^d$ as $\xi \mapsto \mathbf{s} + (\widehat{\mathbf{r} - \mathbf{s}})\xi$. Then the ray transform in (2.1) can be expressed in these coordinates as follows.

Definition 5.1 (Ray transform). *The ray transform of a function $\rho: \mathbb{R}^d \rightarrow \mathbb{R}$ evaluated at a line given by the source-detector pair $(\mathbf{s}, \mathbf{r}) \in \mathbb{R}^d \times \mathbb{R}^d$ is the line integral of ρ over that line:*

$$(5.1) \quad \mathcal{X}(f(\cdot, t))[\mathbf{s}, \mathbf{r}] := \int_{\mathbb{R}} f(\mathbf{s} + (\widehat{\mathbf{r} - \mathbf{s}})\xi, t) d\xi,$$

where $\widehat{\mathbf{x}} := \mathbf{x}/\|\mathbf{x}\|$ for any $\mathbf{x} \in \mathbb{R}^d \setminus \{0\}$, and $\|\cdot\|$ is the d -dimensional Euclidean norm.

Proposition 5.2 (Ray transform of an anisotropic Gaussian). *Let $A = (U^{-1}, \mu) \in \text{Aff}(d)$. Then the ray transform of the anisotropic Gaussian function $A \cdot \rho_0$ is*

$$(5.2) \quad \mathcal{X}(A \cdot \rho_0)[\mathbf{s}, \mathbf{r}] = \frac{\sqrt{\pi}}{\|U(\widehat{\mathbf{r} - \mathbf{s}})\|} \exp\left(\frac{\langle U(\mathbf{r} - \mathbf{s}), U(\mathbf{s} - \mu) \rangle^2}{\|U(\mathbf{r} - \mathbf{s})\|^2} - \|U(\mathbf{s} - \mu)\|^2\right)$$

for $(\mathbf{s}, \mathbf{r}) \in \mathbb{R}^d \times \mathbb{R}^d$, where $(A \cdot \rho)(x) := \rho(A \cdot x) = \rho(U(x - \mu))$.

Proof. Write $\delta = \mathbf{r} - \mathbf{s}$ for brevity. Then

$$\begin{aligned} \mathcal{X}(A \cdot \rho_0)[\mathbf{s}, \mathbf{r}] &= \int_{\mathbb{R}} (A \cdot \rho_0)(\mathbf{s} + \xi \widehat{\delta}) d\xi \\ &= \int_{\mathbb{R}} \exp\left(-\|U(\mathbf{s} + \xi \widehat{\delta} - \mu)\|^2\right) d\xi \\ &= \int_{\mathbb{R}} \exp\left(-\|\xi U \widehat{\delta} + U(\mathbf{s} - \mu)\|^2\right) d\xi. \end{aligned}$$

Expanding the squared norm:

$$(5.3) \quad \mathcal{X}(A \cdot \rho_0)[\mathbf{s}, \mathbf{r}] = \int_{\mathbb{R}} \exp\left(-\|U \widehat{\delta}\|^2 \xi^2 - 2\langle U \widehat{\delta}, U(\mathbf{s} - \mu) \rangle \xi - \|U(\mathbf{s} - \mu)\|^2\right) d\xi$$

$$= \exp\left(-\|\mathbf{U}(\mathbf{s} - \boldsymbol{\mu})\|^2\right) \int_{\mathbb{R}} \exp\left(-\|\mathbf{U}\hat{\boldsymbol{\delta}}\|^2 \xi^2 - 2\langle \mathbf{U}\hat{\boldsymbol{\delta}}, \mathbf{U}(\mathbf{s} - \boldsymbol{\mu}) \rangle \xi\right) d\xi.$$

Applying the identity $\int_{\mathbb{R}} e^{-at^2+bt} dt = \sqrt{\pi/a} e^{b^2/4a}$ (valid for $a > 0$) with $a = \|\mathbf{U}\hat{\boldsymbol{\delta}}\|^2$ and $b = -2\langle \mathbf{U}\hat{\boldsymbol{\delta}}, \mathbf{U}(\mathbf{s} - \boldsymbol{\mu}) \rangle$:

$$\mathcal{X}(\mathbf{A}, \rho_0)[\mathbf{s}, \mathbf{r}] = \exp\left(-\|\mathbf{U}(\mathbf{s} - \boldsymbol{\mu})\|^2\right) \cdot \frac{\sqrt{\pi}}{\|\mathbf{U}\hat{\boldsymbol{\delta}}\|} \exp\left(\frac{\langle \mathbf{U}\hat{\boldsymbol{\delta}}, \mathbf{U}(\mathbf{s} - \boldsymbol{\mu}) \rangle^2}{\|\mathbf{U}\hat{\boldsymbol{\delta}}\|^2}\right).$$

Since $\langle \mathbf{U}\hat{\boldsymbol{\delta}}, \mathbf{U}(\mathbf{s} - \boldsymbol{\mu}) \rangle / \|\mathbf{U}\hat{\boldsymbol{\delta}}\|^2 = \langle \mathbf{U}\boldsymbol{\delta}, \mathbf{U}(\mathbf{s} - \boldsymbol{\mu}) \rangle / \|\mathbf{U}\boldsymbol{\delta}\|^2$, substituting $\boldsymbol{\delta} = \mathbf{r} - \mathbf{s}$ back gives (5.2). ■

The identity in (5.2) allows us to express the forward operator in (4.4) when the lines in the sinogram are parameterized by source–detector pairs:

$$(5.3) \quad F[\Gamma](\mathbf{s}, \mathbf{r}, t) := \sum_{n=1}^N \alpha_n \mathcal{X}\left(\mathbf{A}[\mathbf{U}_n, \boldsymbol{\Theta}_n, \boldsymbol{\eta}_n](t) \cdot \rho_0\right)[\mathbf{s}, \mathbf{r}].$$

In an experimental setup, we only measure finitely many lines at finitely many time samples. A straightforward formalization of this setting is to consider finitely many triplets indexed by the multi-index $\mathbf{m} \in \mathbb{I}$, where $\mathbb{I} = \{1, \dots, M_s\} \times \{1, \dots, M_r\} \times \{1, \dots, M_t\}$:

$$(5.4) \quad \mathbb{W}_0 = \{(\mathbf{s}_{m_s}, \mathbf{r}_{m_r}, t_{m_t})\}_{\mathbf{m} \in \mathbb{I}} \subset \mathbb{W} := \mathbb{M} \times [0, t_{\max}],$$

each encoding the line between a source–detector pair $(\mathbf{s}_m, \mathbf{r}_m)$ at some time point t_m .

Recall from Section 4 that the domain of F consists of elements $\Gamma := (\gamma_1, \dots, \gamma_N)$ of the form

$$(5.5) \quad \gamma_n := ((\alpha_n, \mathbf{U}_n), \boldsymbol{\Theta}_n, \boldsymbol{\eta}_n) \in (\mathbb{R} \times \mathbb{R}^{d_U}) \times \mathbb{R}^{d_R} \times (\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d) = \mathbb{D}.$$

Each element in $\mathbb{D}$ specifies a particle morphology and dynamics. Note that, by Remark 1 the dimension of $\mathbf{U}_n$ is $d(d+1)/2$, so the dimension of $\mathbb{D}$ is $1 + d(d+1)/2 + d + d(d-1)/2 + d + d = d^2 + 3d + 1$. Consequently, the dimension of the domain is $(d_U + d_R + 3d + 1) \times N = (d^2 + 3d + 1) \times N$, with N denoting the number of particles that make up the object. On the other hand the dimension of the measured data is $(d + d + 1) \times M = (2d + 1) \times M$. For fixed number of sources, M_s , and detector elements M_r , one has $M = M_s \times M_r \times M_t$, where M_t is the number of time samples. As a result, M_t should be chosen to at least satisfy

$$(5.6) \quad (2d + 1) \times M \geq (d^2 + 3d + 1) \times N \implies M_s \times M_r \times M_t \geq \frac{(d^2 + 3d + 1)}{2d + 1} \times N$$

Note however that the above inequality does not guarantee that the digitized ray transform in (5.1) restricted to the manifold (5.4) is invertible. To see this, consider a planar setting, i.e. $d = 2$, with a single source position and a line detector of 512 detector elements (receivers). Then (5.6) becomes

$$M_t \geq 11/5 \times N/512 > 0$$

which for $N = 1$ means that $M_t \geq 1$ which is reconstruction from a single view. However, for $M_t = 1$ the inversion problem reduces to the standard tomography problem and in that setting, it is well known that an arbitrary object cannot be recovered from a single view. It is thus clear that one needs additional requirements to ensure invertibility. In addition, further requirements will be needed to ensure the inversion is mildly ill-posed or even better, well-posed.

6. The inverse problem. We now revisit the spatiotemporal inverse problem in Definition 2.1. Rephrasing it in the setting defined by the specific parameterizations introduced in Sections 3–4 gives us the following inverse problem:

Definition 6.1 (The inverse problem). *Recover the parameters governing the motion and morphology for N distinct particles $\Gamma := (\gamma_1, \dots, \gamma_N)$ with $\gamma_i \in \mathbb{D}$ from data $\{g[\mathbf{m}]\}_{\mathbf{m} \in \mathbb{I}} \subset \mathbb{R}$, where*

$$g[\mathbf{m}] \approx F[\Gamma](\mathbf{s}_{m_s}, \mathbf{r}_{m_r}, t_{m_t})$$

for each triplet $(\mathbf{s}_{m_s}, \mathbf{r}_{m_r}, t_{m_t})$ indexed by the multi-index $\mathbf{m} = (m_s, m_r, m_t) \in \mathbb{I}$. The index set $\mathbb{I} = \{1, \dots, M_s\} \times \{1, \dots, M_r\} \times \{1, \dots, M_t\}$ defines the finite sampling of the manifold $\mathbb{M} \times [0, t_{\max}]$, and the forward operator F models the observation process as defined in (5.3).

Least-squares type solutions of the inverse problem in Definition 6.1 are obtained by solving

$$(6.1) \quad \hat{\Gamma} \in \arg \min_{\Gamma} \mathcal{L}(\{g[\mathbf{m}]\}_{\mathbf{m} \in \mathbb{I}}, \{F[\Gamma](\mathbf{s}_{m_s}, \mathbf{r}_{m_r}, t_{m_t})\}_{\mathbf{m} \in \mathbb{I}})$$

where $\mathcal{L}: \mathbb{R}^{|\mathbb{I}|} \times \mathbb{R}^{|\mathbb{I}|} \rightarrow \mathbb{R}$ is some suitably chosen loss function and $|\mathbb{I}| = M_s \times M_r \times M_t$. For example, choosing $\mathcal{L}$ as a p -norm yields

$$(6.2) \quad \hat{\Gamma} \in \arg \min_{\Gamma} \left(\sum_{\mathbf{m} \in \mathbb{I}} |g[\mathbf{m}] - F[\Gamma](\mathbf{s}_{m_s}, \mathbf{r}_{m_r}, t_{m_t})|^p \right)^{1/p}.$$

The spatiotemporal inverse problem in Definition 2.1 is ill-posed in that general setting since inverting the ray transform is known to be unstable. This means in particular that least-squares solutions are unsuitable, as such solution methods are unstable w.r.t. perturbations/errors in data. However, the specific variant of the inverse problem in Definition 6.1 may come with better stability properties (the specific parameterization introduced in Sections 3–4 will here implicitly act as a regularization). This is also why a least-squares solution, like the one in (6.2) for $p = 2$, may work without any additional regularizer.

Besides stability there is the issue of uniqueness. The least-squares optimization problem is non-convex, meaning that it can have several optimal solutions. In addition, gradient-based schemes for solving it could get trapped in undesirable local minima, so initialization becomes important. One approach for mitigating the risk of getting trapped in undesirable local minima is to split the updating of the dynamic motion parameters from the static morphological parameters.

6.1. Intertwined recovery of motion and morphology. The control variable (the object we seek to recover) can be split as $\Gamma = [\Gamma_D, \Gamma_S]$ where $\Gamma_D = [\Gamma_P, \Gamma_R]$ represents the dynamic motion parameters with Γ_P and Γ_R denoting the projectile and rotational components respectively. Hence,

$$\Gamma_P := \{\boldsymbol{\eta}_n\}_{n=1}^N \in [\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d]^N \quad \text{and} \quad \Gamma_R := \{\boldsymbol{\Theta}_n\}_{n=1}^N \in [\mathbb{R}^{d_R}]^N \quad \text{with } d_R := d(d-1)/2.$$

Furthermore, Γ_S represents the static morphological parameters, so

$$\Gamma_S := \{(\alpha_n, \mathbf{U}_n)\}_{n=1}^N \in [\mathbb{R} \times \mathbb{R}^{d_U}]^N \quad \text{with } d_U := d(d+1)/2.$$

One can now adopt an intertwined iterative approach for computing an approximation to the least-squares solution that alternates updates for the projectile motion parameters

403 Γ_P and the rotational and static morphological parameters $[\Gamma_R, \Gamma_S]$. Translating this into
 404 formulas gives the following intertwined scheme:

$$405 \quad (6.3) \quad \hat{\Gamma}_P^i \in \arg \min_{\Gamma_P} \mathcal{L}_P \left(\{g[\mathbf{m}]\}_{\mathbf{m} \in \mathbb{I}}, \{F[\Gamma_P, \hat{\Gamma}_R^{i-1}, \hat{\Gamma}_S^{i-1}](\mathbf{s}_{m_s}, \mathbf{r}_{m_r}, t_{m_t})\}_{\mathbf{m} \in \mathbb{I}} \right)$$

$$406 \quad (6.4) \quad [\hat{\Gamma}_R^i, \hat{\Gamma}_S^i] \in \arg \min_{[\Gamma_R, \Gamma_S]} \mathcal{L}_S \left(\{g[\mathbf{m}]\}_{\mathbf{m} \in \mathbb{I}}, \{F[\hat{\Gamma}_P^i, \Gamma_R, \Gamma_S](\mathbf{s}_{m_s}, \mathbf{r}_{m_r}, t_{m_t})\}_{\mathbf{m} \in \mathbb{I}} \right)$$

407 where $\mathcal{L}_P: \mathbb{R}^{\mathbb{I}} \times \mathbb{R}^{\mathbb{I}} \rightarrow \mathbb{R}$ and $\mathcal{L}_S: \mathbb{R}^{\mathbb{I}} \times \mathbb{R}^{\mathbb{I}} \rightarrow \mathbb{R}$ are suitably chosen loss functions.
 408 Such an approach may not necessarily converge to minimizers of the objectives, however,
 409 it could be sufficient for certain applications. Furthermore, the result could constitute a
 410 good initialization for future approaches. A key step will be to select $\mathcal{L}_P$ and $\mathcal{L}_S$. A natural
 411 criterion is that these are appropriate quantifiers of similarity of motion and morphological
 412 parameters. Another aspect is computational. As we shall see next, an appropriate choice
 413 of $\mathcal{L}_P$ will allow us to decouple the two optimization problems (6.3) and (6.4). This means
 414 the intertwined approach in (6.3)–(6.4) can be replaced by a sequential approach.

415 **6.2. Sequential recovery of motion and morphology.** It is desirable to choose $\mathcal{L}_P$
 416 so that the objective in (6.3) is independent of the morphological parameters $\hat{\Gamma}_S^{i-1} \in$
 417 $[\mathbb{R} \times \mathbb{R}^{d_U}]^N$. The intertwined scheme in (6.3)–(6.4) that updates projectile motion and
 418 rotation/morphological parameters in an alternating manner can then be replaced by a
 419 sequential scheme. Under this scheme, the projectile motion parameters are independently
 420 estimated in a preprocessing step by solving (6.3).

421 To choose such an $\mathcal{L}_P$, we begin by characterizing how the ray transform maps the
 422 modes of a function (2.9) representing the object (the modes are the points where the func-
 423 tion attains local maxima). Since the ray transform is linear, it is enough to characterize
 424 how the ray transform maps the modes of an anisotropic Gaussian.

425 **Proposition 6.2 (Mode of ray transform of an anisotropic Gaussian).** *Let ρ_0 be an iso-*
 426 *otropic Gaussian, $\mathbf{A} = (\mathbf{U}^{-1}, \boldsymbol{\mu}) \in \text{Aff}(d)$, and $\mathbf{s}$ is a fixed source. The mode of the ray*
 427 *transform $\mathbf{r} \mapsto \mathcal{X}(\mathbf{A}, \rho_0)[\mathbf{s}, \mathbf{r}]$ w.r.t. the detector location is then co-linear with the source*
 428 *and the mode $\boldsymbol{\mu}$ of the anisotropic Gaussian $\mathbf{A}, \rho_0$. Consequently, these arguments are*
 429 *independent of the morphology $\mathbf{U}$ of the anisotropic Gaussian $\mathbf{A}, \rho_0$.*

430 **Proof.** Note first that $\mathbf{A}, \rho_0$ is an anisotropic Gaussian. Then, rewriting (5.2) yields
 431 the following analytic expression for $\mathbf{r} \mapsto \mathcal{X}(\mathbf{A}, \rho_0)[\mathbf{s}, \mathbf{r}]$:

$$432 \quad (6.5) \quad \mathcal{X}(\mathbf{A}, \rho_0)[\mathbf{s}, \mathbf{r}] = \frac{\sqrt{\pi}}{\|\mathbf{U}(\widehat{\mathbf{r} - \mathbf{s}})\|} \exp \left(-\|\mathbf{U}(\mathbf{s} - \boldsymbol{\mu})\|^2 \left[1 - \left\langle \frac{\mathbf{U}(\mathbf{s} - \mathbf{r})}{\|\mathbf{U}(\mathbf{s} - \mathbf{r})\|}, \frac{\mathbf{U}(\mathbf{s} - \boldsymbol{\mu})}{\|\mathbf{U}(\mathbf{s} - \boldsymbol{\mu})\|} \right\rangle^2 \right] \right).$$

433 We now see that $\mathbf{r} \mapsto \mathcal{X}(\mathbf{A}, \rho_0)[\mathbf{s}, \mathbf{r}]$ is bounded above and below by a constant times the
 434 exponential term. Consequently, it obtains its maximum when

$$435 \quad (6.6) \quad 1 - \left\langle \frac{\mathbf{U}(\mathbf{s} - \mathbf{r})}{\|\mathbf{U}(\mathbf{s} - \mathbf{r})\|}, \frac{\mathbf{U}(\mathbf{s} - \boldsymbol{\mu})}{\|\mathbf{U}(\mathbf{s} - \boldsymbol{\mu})\|} \right\rangle^2 = 0 \iff \left| \frac{\mathbf{U}(\mathbf{s} - \mathbf{r})}{\|\mathbf{U}(\mathbf{s} - \mathbf{r})\|} \right| = \left| \frac{\mathbf{U}(\mathbf{s} - \boldsymbol{\mu})}{\|\mathbf{U}(\mathbf{s} - \boldsymbol{\mu})\|} \right|.$$

436 This implies that

$$437 \quad (6.7) \quad \arg \max_{\mathbf{r}} \mathcal{X}(\mathbf{A}, \rho_0)[\mathbf{s}, \mathbf{r}] = \mathbf{s} - t(\mathbf{s} - \boldsymbol{\mu}), \quad \text{for some } t \in \mathbb{R},$$

438 which is independent of the morphological parameter $\mathbf{U}$. ■

439 **Corollary 6.3.** *Assume that the trajectories $t \mapsto \mathbf{C}[\boldsymbol{\eta}_n](t)$ of the $n = 1, \dots, N$ particles*
 440 *lie between the fixed source $\mathbf{s}$ and any of the detectors $\mathbf{r}$. For each t , the modes of $\mathbf{r} \mapsto$*

441 $F[\Gamma](\mathbf{s}, \mathbf{r}, t)$ are then within the neighborhood of the modes of $\mathbf{r} \mapsto \mathcal{X}(A[\mathbf{U}_n, \boldsymbol{\eta}_n](t) \cdot \rho_0)[\mathbf{s}, \mathbf{r}]$.
 442 Furthermore, these t -dependent modes are in turn given by the set of curves

$$443 \quad (6.8) \quad \left\{ t \mapsto (\mathbf{r}, t) \text{ such that } \frac{\mathbf{s} - \mathbf{C}[\boldsymbol{\eta}_n](t)}{\|\mathbf{s} - \mathbf{C}[\boldsymbol{\eta}_n](t)\|} \cdot \frac{\mathbf{s} - \mathbf{r}}{\|\mathbf{s} - \mathbf{r}\|} = 1 \right\}_{n=1}^N.$$

444 *Proof.* The proof is similar to that of Theorem 3.4 in [54] and it is a direct consequence
 445 of Proposition 6.2. ■

446 If one chooses L_P such that the objective in (6.3) corresponds to a distance between
 447 local maxima of $\{g_m\}_{m=1}^M$ and modes of $\mathbf{r} \mapsto \mathcal{X}(A[\mathbf{U}_n, \boldsymbol{\eta}_n](t) \cdot \rho_0)[\mathbf{s}, \mathbf{r}]$, then Proposition 6.2
 448 and Corollary 6.3 gives that the minimizer $\hat{\Gamma}_P^i$ in (6.3) is independent of $\hat{\Gamma}_S^{i-1}$. In particular,
 449 $\hat{\Gamma}_P^i$ will not depend of the morphology, so consequently there is no need for iterating over
 450 i in (6.3)–(6.4). It is therefore advantageous to work with such an L_P as it helps towards
 451 estimating the trajectories of the particles $\mathbf{C}[\boldsymbol{\eta}_n](t)$ without requiring the knowledge of
 452 their rotational motion or morphologies. Consequently, recovery of the remaining rotation
 453 and morphology requires modification of the optimization problem (6.4) to

$$454 \quad (6.9) \quad [\hat{\Gamma}_R, \hat{\Gamma}_S] \in \arg \min_{[\Gamma_R, \Gamma_S]} L_S \left(\{g[\mathbf{m}]\}_{\mathbf{m} \in \mathbb{I}}, \{F[\Gamma_P, \Gamma_R, \Gamma_S](\mathbf{s}_m, \mathbf{r}_m, t_m)\}_{\mathbf{m} \in \mathbb{I}} \right)$$

455 (i.e. dropping the dependency on the sequencing superscript i), where all trajectory
 456 parameters in Γ_P are obtained by solving (6.3) with the aforementioned L_P .

457 In the following we show how to select L_P (given by (6.15)), in the setting with fixed
 458 acquisition geometry and a single source.

459 **6.3. Case with fixed acquisition geometry and a single source.** We here consider
 460 a fixed acquisition geometry with a single source $\mathbf{s}$ and multiple detectors. Using the
 461 multi-index notation $\mathbf{m} = [m_r, m_t]$, we can denote measurements as

$$462 \quad (6.10) \quad g[\mathbf{m}] = g[m_r, m_t] \approx F[\Gamma](\mathbf{s}, \mathbf{r}_{m_r}, t_{m_t}) \quad \text{with } (\mathbf{s}, \mathbf{r}_{m_r}, t_{m_t}) \in \mathbb{M} \times [0, t_{\max}],$$

463 where

$$464 \quad (6.11) \quad \mathbb{I} = \{\mathbf{m} = [m_r, m_t] \text{ such that } m_r = 1, \dots, M_r \text{ and } m_t = 1, \dots, M_t\}$$

465 is the set of detector and time indices.

466 In this setting, we optimize using the observed modes present in the projection data,
 467 and the predicted modes that can be obtained as a function of a trajectory parameter
 468 prediction $\boldsymbol{\eta}$.

469 **Definition 6.4.** Let $M[g](t)$ denote the points of local maxima of $g[\mathbf{m}]$ with respect to a
 470 fixed arrangement of detectors $\mathbf{r} \in \{\mathbf{r}_{m_r}\}_{m_r=1}^{M_r}$ and a fixed set of time points $t \in \{t_{m_t}\}_{m_t=1}^{M_t}$.
 471 Furthermore, define

$$472 \quad C_{\mathcal{X}}(t) := \bigcup_{n=1}^N C_{\mathcal{X}}[\boldsymbol{\eta}_n](t),$$

473 where

$$474 \quad (6.12) \quad C_{\mathcal{X}}[\boldsymbol{\eta}_n](t) = \arg \min_{\mathbf{r}} \left| \frac{\mathbf{s} - \mathbf{C}[\boldsymbol{\eta}_n](t)}{\|\mathbf{s} - \mathbf{C}[\boldsymbol{\eta}_n](t)\|} \cdot \frac{\mathbf{s} - \mathbf{r}}{\|\mathbf{s} - \mathbf{r}\|} - 1 \right|$$

475 and where $\mathbf{r}$ in (6.12) is considered over the subset of $\mathbb{M}$ that describes the detector domain.

Equation (6.12) provides an expression for the theoretical mode of the n -th Gaussian projection, but it is not an explicit map from trajectory parameter space to mode space and is therefore not applicable to gradient descent optimization. Instead, we require the following equivalent result.

Proposition 6.5. *Let $d \geq 2$, let $\boldsymbol{\eta}_n \subset \Gamma_P$, and let $\mathbf{C}[\boldsymbol{\eta}_n](t)$ be the corresponding $\mathbb{R}^d$ -valued trajectory. The unique mode of $\alpha_n \mathcal{X}(\mathcal{A}[\mathbf{U}_n, \boldsymbol{\eta}_n](t) \cdot \rho_0)[s, r]$ is*

$$(6.13) \quad \hat{\mathbf{r}}[\boldsymbol{\eta}_n](t) = \mathbf{s} + \lambda_n (\mathbf{s} - \mathbf{C}[\boldsymbol{\eta}_n](t)),$$

where

$$\lambda_n = \frac{\mathbf{r}^{(0)} - \mathbf{s}^{(0)}}{\mathbf{s}^{(0)} - \mathbf{C}^{(0)}[\boldsymbol{\eta}_n](t)}$$

and $\mathbf{v}^{(0)}$ denotes a fixed component of $\mathbf{v}$. In particular, (6.13) defines an explicit smooth map from trajectory parameter space to mode space.

Proof. By Corollary 6.3 the mode is morphology independent, so it suffices to take $\mathbf{U}_n = \xi \mathbf{I}$, $\xi > 0$. Setting $\tilde{\mathbf{R}}_{\Omega_n}(t) := \mathbf{U}_n \mathbf{R}_{\Omega_n}^\top(t)$, one computes

$$\|\tilde{\mathbf{R}}_{\Omega_n}(t)(\mathbf{r} - \mathbf{s})\| = \xi \|\mathbf{r} - \mathbf{s}\|, \quad \langle \tilde{\mathbf{R}}_{\Omega_n}(t)(\mathbf{r} - \mathbf{s}), \tilde{\mathbf{R}}_{\Omega_n}(t)(\mathbf{s} - \mathbf{C}) \rangle = \xi^2 \langle \mathbf{r} - \mathbf{s}, \mathbf{s} - \mathbf{C} \rangle,$$

where $\mathbf{C} = \mathbf{C}[\boldsymbol{\eta}_n](t)$. Accordingly, the ray transform factors as $\mathcal{X}(\mathcal{A}[\mathbf{U}_n, \boldsymbol{\eta}_n](t) \cdot \rho_0)[\mathbf{s}, \mathbf{r}] = P_n(\mathbf{s}, t) Q_n(\mathbf{s}, \mathbf{r}, t)$ with

$$P_n := \sqrt{\pi} \exp(-\|\tilde{\mathbf{R}}_{\Omega_n}(t)(\mathbf{s} - \mathbf{C})\|^2), \quad Q_n := \exp\left(\xi^2 \frac{\langle \mathbf{r} - \mathbf{s}, \mathbf{s} - \mathbf{C} \rangle^2}{\|\mathbf{r} - \mathbf{s}\|^2}\right).$$

Since P_n is $\mathbf{r}$ -independent, modes are determined by $\nabla_{\mathbf{r}} Q_n = \mathbf{0}$. Computing:

$$\frac{\partial Q_n}{\partial r_i} = \frac{2Q_n}{\|\mathbf{r} - \mathbf{s}\|^4} \langle \mathbf{r} - \mathbf{s}, \mathbf{s} - \mathbf{C} \rangle \left(\|\mathbf{r} - \mathbf{s}\|^2 (\mathbf{s} - \mathbf{C})_i - (\mathbf{r} - \mathbf{s})_i \langle \mathbf{r} - \mathbf{s}, \mathbf{s} - \mathbf{C} \rangle \right).$$

The prefactor $\langle \mathbf{r} - \mathbf{s}, \mathbf{s} - \mathbf{C} \rangle = 0$ would require $\mathbf{r} - \mathbf{s} \perp \mathbf{s} - \mathbf{C}$, which is geometrically excluded in the acquisition geometry. Hence $\nabla_{\mathbf{r}} Q_n = \mathbf{0}$ reduces to

$$(6.14) \quad \|\mathbf{u}\|^2 \mathbf{v}_n - \langle \mathbf{u}, \mathbf{v}_n \rangle \mathbf{u} = 0, \quad \mathbf{u} := \mathbf{r} - \mathbf{s}, \quad \mathbf{v}_n := \mathbf{s} - \mathbf{C}[\boldsymbol{\eta}_n](t).$$

For $\mathbf{u}, \mathbf{v}_n \neq \mathbf{0}$, (6.14) holds if and only if $\mathbf{u}$ and $\mathbf{v}_n$ are parallel, giving $\mathbf{r} = \mathbf{s} + \lambda(\mathbf{s} - \mathbf{C}[\boldsymbol{\eta}_n](t))$. Fixing λ via the known component data $\mathbf{s}^{(0)}, \mathbf{r}^{(0)}, \mathbf{C}^{(0)}[\boldsymbol{\eta}_n](t)$ yields (6.13). ■

Proposition 6.2 states that the projection of the trajectories of Gaussian particles within a GMM object should be in the vicinity of the modes of the ray transform of the object. Based on this, we select $\mathbf{L}_P$ by minimizing the Hausdorff distance between $\mathbf{M}[g](t)$ and $\{\hat{\mathbf{r}}[\boldsymbol{\eta}_n](t)\}_{n=1}^N$ over all time samples:

$$(6.15) \quad \mathbf{L}_P\left(\{g[\mathbf{m}]\}_{\mathbf{m} \in \mathbb{I}}, \{F[\Gamma_P, \Gamma_R, \Gamma_S](\mathbf{s}, \mathbf{r}_{m_r}, t_{m_t})\}_{\mathbf{m} \in \mathbb{I}}\right) := \sum_{m_t=1}^{M_t} \sum_{n=1}^N d_H\left(\mathbf{M}[g](t_{m_t}), \hat{\mathbf{r}}[\boldsymbol{\eta}_n](t_{m_t})\right)$$

where d_H is the Hausdorff distance between subsets of $\mathbb{R}^d$:

$$(6.16) \quad d_H(\mathbb{A}, \mathbb{B}) = \max\left(\sup_{\mathbf{a} \in \mathbb{A}} \inf_{\mathbf{b} \in \mathbb{B}} \|\mathbf{a} - \mathbf{b}\|, \sup_{\mathbf{b} \in \mathbb{B}} \inf_{\mathbf{a} \in \mathbb{A}} \|\mathbf{b} - \mathbf{a}\|\right) \quad \text{for } \mathbb{A}, \mathbb{B} \subset \mathbb{R}^d.$$

Consequently, the optimal trajectory parameters are given by

$$(6.17) \quad \Gamma_P^* = \{\boldsymbol{\eta}_n^*\}_{n=1}^N = \arg \min_{\{\boldsymbol{\eta}_n\}_{n=1}^N} \sum_{m_t=1}^{M_t} \left[\sum_{n=1}^N d_H\left(\mathbf{M}[g](t_{m_t}), \hat{\mathbf{r}}[\boldsymbol{\eta}_n](t_{m_t})\right) \right].$$

Figures 8.3–8.4 visualize the pre and post-optimization mode fitting procedure for a two-dimensional numerical example. Note that computing the optimal trajectory parameters $\{\eta_n^*\}_{n=1}^N$ in (6.17) amounts to the following: We assign the modes of the projected particles (that make up the object) to the modes of the data in a way that minimizes the total error over all time points t_{m_t} . Computing exact Hausdorff distances can be computationally expensive, especially for high-dimensional imprecise data [45, 23], but there are efficient approximate methods [10, 17].

After approximately solving (6.3) via (6.17), the trajectories are fixed, and it remains to recover the rotation and morphology parameters as described by (6.4).

In the case of trajectory optimization, formulating (6.3) in terms of the modes serves to provide a concise measure of the projectile motion while being invariant with respect to morphology and rotational motion. In contrast, because the (α_n) determine how the projection integral is scaled across all acquisition times, and the (U_n) determine the spatial extent as a function of the viewing angle, the projection data must be utilized in its entirety when fitting for the particle morphology (α_n, U_n) and angular velocity (Θ_n) .

Consequently, L_S is chosen to directly measure the discrepancy between the measured and modeled projections. One option is to choose L_S as the L_p norm for $p \geq 1$ or some (smooth) approximation it. We consider a smooth differentiable approximation of the L_1 – the Huber loss function [21] over the entirety of the sinogram:

$$(6.18) \quad L_S\left(\{g[\mathbf{m}]\}_{\mathbf{m}}, \{F[\Gamma_P, \Gamma_R, \Gamma_S](\mathbf{s}, \mathbf{r}_{m_r}, t_{m_t})\}_{m_r=1, m_t=1}^{M_r, M_t}\right) \\ := \frac{1}{M_t \times M_r} \sum_{\mathbf{m}} H_\delta\left(g[\mathbf{m}] - F[\Gamma_P, \Gamma_R, \Gamma_S](\mathbf{s}, \mathbf{r}_{m_r}, t_{m_t})\right)$$

where

$$(6.19) \quad H_\delta(u) := \begin{cases} \frac{1}{2}u^2 & |u| \leq \delta, \\ \delta(|u| - \frac{\delta}{2}) & |u| > \delta. \end{cases}$$

Consequently,

$$(6.20) \quad [\Gamma_R^*, \Gamma_S^*] = \arg \min_{[\Gamma_R, \Gamma_S]} \frac{1}{M_t \times M_r} \sum_{\mathbf{m}} H_\delta\left(g[\mathbf{m}] - F[\Gamma_P^*, \Gamma_R, \Gamma_S](\mathbf{s}, \mathbf{r}_{m_r}, t_{m_t})\right)$$

where Γ_P^* are the optimal projectile motion parameters given by (6.17).

7. Reconstruction algorithm.

7.1. Initialization. The manner in which the parameter estimates are initialized reflects the multi-stage nature of the algorithm.

7.1.1. Stage 1 initialization: Trajectories. In keeping with (6.3)–(6.4), the first initialization to consider is for the trajectory optimization step. The initial estimates are computed as independent and identically distributed (i.i.d.) samples from the probability distribution

$$(7.1) \quad \eta_n \sim \mathcal{N}(\boldsymbol{\mu}_{\text{traj}}, \boldsymbol{\Sigma}_{\text{traj}}), \quad n = 1 \dots, N,$$

for some specified mean vector and covariance matrix $\boldsymbol{\mu}_{\text{traj}}$ and $\boldsymbol{\Sigma}_{\text{traj}}$ respectively. Once sampled, the N trajectory vectors are then optimized by solving (6.17).

To perform the optimization, placeholders for the $(\alpha_n, U_n, \Theta_n)$ are required. Recalling the independence of rotation and morphology from projectile motion under (6.15), these

parameters can be set to any value, provided $(\mathbf{U}_n^\top \mathbf{U}_n)^{-1}$ is symmetric positive definite. However, for the subsequent angular velocity initialization it is convenient here to choose the $(\mathbf{U}_n)$ to produce anisotropic Gaussian particles, with axis scales commensurate with the physical dimensions of the problem. An exact choice of values of the placeholders for $(\alpha_n, \boldsymbol{\Theta}_n, \mathbf{U}_n)$, $\boldsymbol{\mu}_{\text{traj}}$ and $\boldsymbol{\Sigma}_{\text{traj}}$ for a 2D example are given in Section 8.

To avoid settling on an optimized trajectory solution that is only a local minimum, the random sampling procedure (7.1) is repeated N_{traj} times. The definitive optimized trajectory solution is then chosen as the one with the minimum terminated loss function value. Such an approach is practically feasible due to the per-time $\mathcal{O}(Nd)$ evaluation cost of the GMM trajectory forward model. Furthermore, accurate trajectory estimation is fundamental to the quality of the overall reconstruction; in the absence of a data-informed or deterministic alternative to (7.1), repeated trials are highly worthwhile.

7.1.2. Stage 2 initialization: Rotation and attenuation. Post-trajectory optimization, the projectile motion of each particle is known and fixed. Consequently, the sinogram contribution of any particle subset conditioned on the placeholders $(\alpha_n, \boldsymbol{\Theta}_n, \mathbf{U}_n)$ is known and can be held constant, allowing the residual to isolate information about the remaining particles.

Exploiting this property, the initialized rotation for the n -th particle ρ_n is defined as that which minimizes the residual sinogram; i.e.

$$(7.2) \quad \arg \min_{\boldsymbol{\Theta}_n} \left\| g[\mathbf{m}] - \sum_{\ell \neq n} F[\gamma_\ell](\mathbf{s}, \mathbf{r}_{m_r}, t_{m_t}) - F[\gamma_n(\boldsymbol{\Theta}_n)](\mathbf{s}, \mathbf{r}_{m_r}, t_{m_t}) \right\|_{\ell^2}, \quad n = 1, \dots, N.$$

Note that (7.2) only optimizes over $\boldsymbol{\Theta}_n$ — all other parameters are fixed for ρ_n , and all parameters are fixed for all other ρ_ℓ , $\ell \neq n$. A necessary condition for $\boldsymbol{\Theta}_n$ to be identifiable from (7.2) is that the ρ_n are anisotropic (isotropic ρ_n would have infinitely many solutions). Provided the $(\mathbf{U}_n)$ have been initialized as such in Stage 1, then all $(\alpha_n, \boldsymbol{\eta}_n, \mathbf{U}_n)$ can remain fixed as they are. While the values of $(\alpha_n, \mathbf{U}_n)$ currently proposed here are naive estimates that introduce bias, they can provide a sufficient heuristic for computing a reasonable initialization in $\boldsymbol{\Theta}_n$.

To approximate (7.2), a grid search strategy over a grid of N_{rot} equally-spaced points within $[\omega_{\min}, \omega_{\max}]$ is performed for each component of $\boldsymbol{\Theta}_n \in \mathbb{R}^{d_R}$. This avoids the local minima issues that compromise a gradient descent strategy in highly oscillatory loss landscapes. While an exhaustive grid search scales as $\mathcal{O}(N_{\text{rot}}^{d_R})$, a coordinate-wise approach reduces the cost to $\mathcal{O}(d_R N_{\text{rot}})$ at the price of missing cross-axis interactions when $d > 2$ (a non-issue when $d = 2$, since then $d_R = 1$). Whether this approximation is sufficiently accurate for initialization in higher dimensions is an empirical question for future investigation.

With the $(\boldsymbol{\Theta}_n)$ initialized, for the attenuation coefficients (α_n) , let $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_N)$. The corresponding initialized values are defined to be the solution of

$$(7.3) \quad \arg \min_{\boldsymbol{\alpha}} \|\Phi \boldsymbol{\alpha} - g[\mathbf{m}]\|_{\ell^2}^2,$$

where the n -th column of Φ is the full sinogram of ρ_n with $\alpha_n = 1$ and $(\boldsymbol{\eta}_n, \mathbf{U}_n, \boldsymbol{\Theta}_n)$ are fixed from the previous stages. To ensure non-negativity, (7.3) is approximately solved via least-squares with non-negativity enforced by projection [27].

As the precision matrices $(\mathbf{U}_n)$ are not directly fit to the projection data prior to Stage 2, the capacity for (7.2) and (7.3) to provide exact initializations is limited. However, these methods can provide an accurate enough starting point for Stage 2 to converge

reliably in a small number of trials N_{morph} . This is desirable due to the relatively high per-time evaluation cost $\mathcal{O}(Nd^2)$ of (6.4).

Lastly, we note that we do not initialize for the number of particles N and assume it to be known, which in certain applications such as subsurface characterization would likely not be the case. This is a task in model order selection [8, 42] which we leave for future work.

7.2. Pseudocode. We summarize the tomographic reconstruction algorithm for GMM objects undergoing parabolic motion. The algorithm, called GMM-CT, is structured as below, an open source implementation of GMM-CT is available at <https://github.com/dwb26/gmm-ct>.

Input:

- Dynamic sinogram: Time series of data $g[\mathbf{m}] = g[m_{\mathbf{r}}, m_t]$.
- Source and detector locations, and time samples: $(\mathbf{s}, \mathbf{r}_{m_{\mathbf{r}}}, t_{m_t}) \in \mathbb{M} \times [0, t_{\max}]$.
- Number of particles: N .

Output:

- Projectile motion parameters of particles: Γ_P^* .
- Rotational motion parameters of particles: Γ_R^* .
- Static morphological parameters of particles: Γ_S^* .

The output is computed following a two-stage procedure:

Stage 1: Estimating motion (projectile trajectory) parameters Γ_P^*

- Estimate modes of the projection data $\{\mathbf{M}[g](t_{m_t})\}_{m_t=1}^{M_t}$.
- Initialization of the projectile motion trajectory parameters $\{\boldsymbol{\eta}_n\}_{n=1}^N$.
- Compute predicted modes of the trajectories $\{\hat{\mathbf{r}}[\boldsymbol{\eta}_n](t_{m_t})\}_{m_t=1}^{M_t}$.
- Estimate optimal trajectory parameters $\Gamma_P^* = \{\boldsymbol{\eta}_n^*\}_{n=1}^N$ by solving (6.17).

Stage 2: Estimating rotational motion and morphological parameters $[\Gamma_R^*, \Gamma_S^*]$

- Initialization of morphological parameters Γ_S .
- Initialization of angular velocity parameters Γ_R via grid search over $[\omega_\ell, \omega_r]$.
- Initialization of attenuation coefficients $\boldsymbol{\alpha}$ via least squares with enforced non-negativity (7.3).
- Estimate optimal rotational motion and morphology parameters $[\Gamma_R^*, \Gamma_S^*]$ by solving (6.20).

To provide an approximation to the Hausdorff distance in (6.17) the Hungarian algorithm is used [25], in particular the SciPy implementation of the Jonker-Volgenant algorithm [15] due to the rectangular nature of the assignment problem. Furthermore, numerical experiments indicate that additional accuracy gains for the trajectories can be achieved using a Newton root-finder to estimate the $\boldsymbol{\eta}_n$ such that $\nabla_{\mathbf{r}} Q_n = \mathbf{0}$ in the proof of Proposition 6.5 — a step that can only be correctly applied once the assignments are established. The accuracy of the trajectory reconstruction is crucial to the success of the remaining reconstruction, making such accuracy gains highly worthwhile.

8. Numerical results. For the rest of our discussion, the units of all spatial variables should be considered in meters (m) and time should be considered in seconds (s).

Modeling a two-dimensional scenario, we consider a single source located at $(-1, -1)^\top$ and a vertical linear detector geometry at $4 \times [-3, 1]$ in $\mathbb{R}^2$ as shown in Figure 8.1. The particles are assumed to be ejected from a known location $(1, 1)^\top$ with unknown velocities, subject to known gravitational acceleration $(0, -9.81)^\top$ (m s^{-2}). Consequently, $\mathbf{C}[\boldsymbol{\eta}_n](t)$ is characterized by its unknown initial velocity $\mathbf{v}_n \in \mathbb{R}^d$.

For $n = 1, \dots, N$, the simulated GMM parameters are generated with respect to the

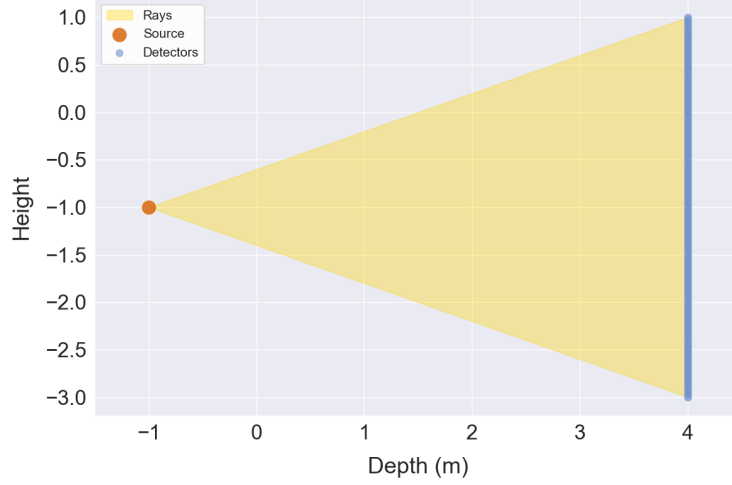


Figure 8.1: Acquisition geometry, with one source located at $(-1, 1)^\top$ and $M_r = 128$ equally-spaced detectors placed between $(4, 1)^\top$ and $(4, -3)^\top$.

639 following models.

- 640 • **Attenuation:** $\alpha_n \sim \mathcal{N}(15 + 5(n - 1), 1^2)$, constrained to $\alpha_n > 0$.
- 641 • **Skewness matrix:** $\mathbf{U}_n \in \mathbb{R}^{d \times d}$ is upper triangular with

$$642 \quad [\mathbf{U}_n]_{ij} \sim \begin{cases} \mathcal{U}(7.5, 25.5) & i = j, \\ \mathcal{N}(10, 1^2) & i < j, \\ 0 & i > j, \end{cases}$$

643 subject to $[\mathbf{U}_n]_{ii} > 0$ and a minimum anisotropy ratio $\max_i [\mathbf{U}_n]_{ii} / \min_i [\mathbf{U}_n]_{ii} \geq \delta_U$
 644 with $\delta_U = 1.5$, enforced by rejection sampling. Near-isotropic Gaussians ($\delta_U \approx 1$)
 645 produce negligible rotation signature in the projections, making Θ_n practically
 646 unidentifiable.

- 647 • **Angular velocity:** $\Theta_n \sim \mathcal{U}(\omega_\ell, \omega_r)$ with $\omega_\ell = 2.0$ and $\omega_r = 6.0$ (rad s^{-1}), subject
 648 to rejection of values for which the angular displacement per projection $\Theta_n \Delta t$ is
 649 within a 10% guard-band of a half-turn multiple, which would render the rotation
 650 undetectable at the given sampling rate.
- 651 • **Initial position and acceleration:** $\mathbf{x}_n = (1, 1)^\top$ and $\mathbf{a}_n = (0, -9.81)^\top$ (m s^{-2}),
 652 shared and known for all n .
- 653 • **Initial velocity:** $\mathbf{v}_n$ (m s^{-1}) is chosen deterministically as listed in Table 8.2,
 654 with horizontal velocity components satisfying $(\mathbf{v}_n)_1 > 0$ to ensure particles travel
 655 towards the detectors within the modeled gantry.

656 Based on samples from these distributions, the corresponding (noiseless) sinogram is
 657 observed and presented in Figure 8.2.

658 For $n = 1, \dots, N$, the models used in the initialization are as follows.

- 659 • **Initial velocity:** The initial i.i.d. sampled velocities for trajectory optimization
 660 are each drawn as

$$661 \quad \mathbf{v}_n \sim \mathcal{N}((1, 1)^\top, 1.5^2 \mathbf{I}_2).$$

- 662 • **Morphology:** The skewness matrices are each initialized according to

$$663 \quad \mathbf{U}_n = \begin{bmatrix} 30 & 0 \\ 0 & 15 \end{bmatrix} + \mathbf{E},$$

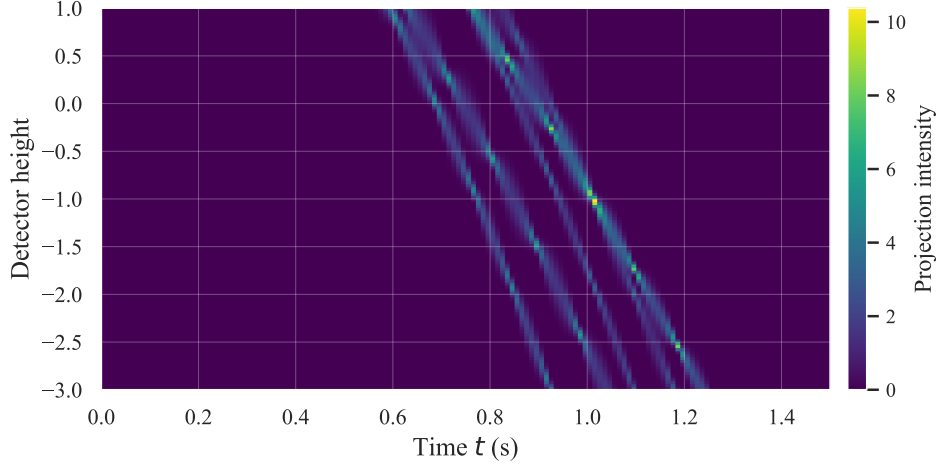


Figure 8.2: Projection data corresponding to $N = 5$ Gaussian particles in projectile motion.

where $\mathbf{E} \in \mathbb{R}^{2 \times 2}$ is strictly upper triangular with entries given by i.i.d. samples from $\mathcal{N}(0, 1)$.

- **Attenuation:** $\alpha_n = 12.5$ for all n ; (7.3) replaces these with least squares with enforced non-negativity estimates.
- **Angular velocity:** $\Theta_n = 0$ for all n . The Stage 2 initialization replaces these via a grid search over a specified mesh grid interval $[\omega_\ell, \omega_r]$ of $N_{\text{rot}} = 200$ equally spaced points, with $\omega_\ell = 2.0$ and $\omega_r = 6.0$ (rad s^{-1}). In Stage 2 optimization, the first trial warm-starts Θ_n from this initialization, while each of the remaining $N_{\text{morph}} - 1$ trials draws $\Theta_n \sim \mathcal{U}(\omega_\ell, \omega_r)$ independently.

All of the remaining experiment hyperparameters are provided in Table 8.1, while the realized sampled parameter values and their corresponding reconstructed estimates are presented in Table 8.2.

Table 8.1: Numerical experiment hyperparameters.

Parameter	Value	Description
M_t	150	Number of projections.
$(t_{\min}, t_{\max})$	(0, 1.5)	Minimum and maximum simulated time window values (s).
M_r	128	Number of detectors.
N_{traj}	20	Number of randomly initialized trajectory recovery trials.
N_{rot}	200	Number of rotation grid search points.
N_{morph}	3	Number of rotation/structure recovery trials.

Given the projection data, its modes at a time instance are inferred using a peak detection algorithm. The detected peaks for times $t = 0.805, 0.906, 1.007$ and 1.107 seconds are color coded and presented in four subplots shown on left hand side of Figure 8.3. These modes — along with the rest of the observed modes of the sinogram — are presented on the right hand side of Figure 8.3.

Once the modes of the sinogram are computed, optimal trajectory parameters are estimated by minimization of (6.17), which we approximately solve using the L-BFGS method [34, Ch. 7]. In Figure 8.4, along with the observed mode data $\{\mathbf{M}[g](t_{m_t})\}_{m_t=1}^{M_t}$,

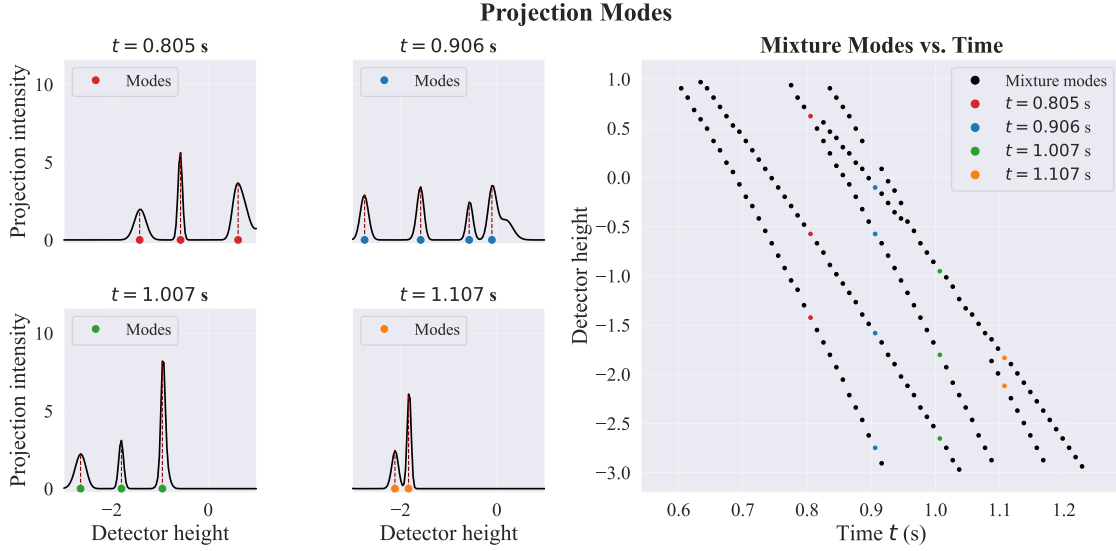


Figure 8.3: Projection data and corresponding color coded estimated modes at times $t = 0.805s$ (red), $t = 0.906s$ (blue), $t = 1.007s$ (green) and $t = 1.107s$ (orange) instances are shown on the small left four subplots. All of the observed modes with the color coded modes on left subplots are shown on the right subplot.

684 the initial sampled and predicted trajectories for the optimal sample are presented in the
 685 left and right panels respectively.

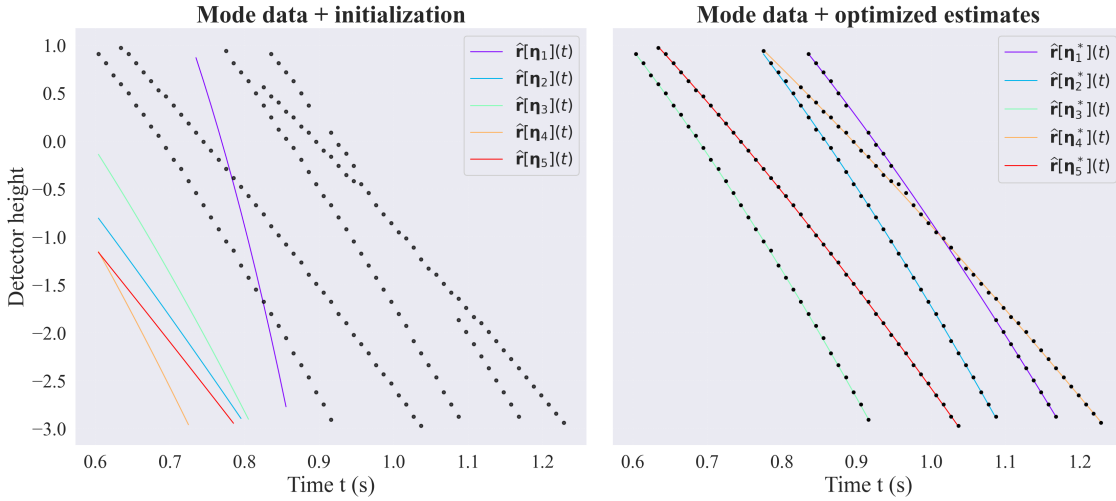


Figure 8.4: Left: The detected modes and the mode maps $\hat{r}[\eta_n]$ resulting from the initial trajectory parameter sample. Right: The mode maps fit to the data to give the optimized trajectory parameter estimates η_n^* , obtained by solving (6.17).

686 The left column plots in both Figure 8.5 and Figure 8.6 visualize the simulated, unob-
 687 served generated Gaussian particles at a collection of sampled projection acquisition times.
 688 The middle column in these figures includes the corresponding samples of the projection
 689 data that inform the reconstruction. The two figures differ only in the estimates plot in
 690 the right column panels and the corresponding projection data in the middle columns.

691 In Figure 8.5, the Stage 2-initialized particles are visualized prior to optimization. The
 692 projection plots in the central column capture the discrepancy in the current reconstruction

state via the difference in projection shape.

In Figure 8.6, the algorithm output demonstrates an accurate object reconstruction, as shown by the resemblance of the right column to the left, informed by an accurate fit in the projection data space in the middle columns.

The projection fit in this case is a sufficient criterion to inform reconstruction in the true, unobserved particle space. However, in general, an accurate projection fit is a necessary but not a sufficient criterion. If particles are sufficiently “unobserved” in the projections — either due to a non-informative rate of angular velocity and/or projection acquisition rate, or due to particles obscuring one another, leading to an identifiability issue — then reconstruction solutions become non-unique. There then exist multiple distinct GMM parameterizations that provide equally valid fits in projection space.

An instance of this obscuring-type behavior can be observed at time $t = 1.01$ in the left columns of Figures 8.5–8.6, where despite the presence of four particles in the acquisition window, only three discernible peaks are observed in the projection data. That the obscured particles are able to be recovered in this example is necessarily partly down to the fact that there exist observation times for which the particles are discernible from one another in the sinogram.

With particle morphology being the primary reconstruction target, what is of most importance from an application perspective is the accuracy of the rigid body reconstruction of the particles. In Figure 8.7 the five simulated, Stage 1-initialized, and reconstructed particles are plotted along the top, second and fourth rows respectively, with the corresponding pairwise $\log_{10}$ absolute errors visualized along the third and bottom rows. The output errors are all approximately in the order of 10^{-2} which, in the context of the simulated Gaussians having average peak amplitudes of 20, represents a relative error of approximately $10^{-2}/20 \approx 0.05\%$ — a substantial improvement on the analogous initial morphology error of approximately $10^0/20 \approx 5\%$. Figure 8.7 is therefore the strongest demonstration of the reconstruction success for this particle batch.

Table 8.2: Ground truth and estimated ($\hat{\cdot}$) parameters for the $N = 5$ Gaussian mixture example, with estimates matched to ground truth via the assigned trajectories. All components share known and fixed initial position $\mathbf{x}_n = (1, 1)^\top$ and gravitational acceleration $\mathbf{a}_n = (0, -9.81)^\top$.

n	Attenuation		Morphology		Rotation		Velocity	
	α_n	$\hat{\alpha}_n$	$\mathbf{U}_n$	$\hat{\mathbf{U}}_n$	Θ_n	$\hat{\Theta}_n$	$\mathbf{v}_n$	$\hat{\mathbf{v}}_n$
1	15.0447	15.1269	$\begin{pmatrix} 10.2751 & 10.3696 \\ 0 & 17.9620 \end{pmatrix}$	$\begin{pmatrix} 8.7032 & 1.5181 \\ 0 & 21.5173 \end{pmatrix}$	5.9297	5.8898	$\begin{pmatrix} 1.00 \\ 3.00 \end{pmatrix}$	$\begin{pmatrix} 0.9965 \\ 2.9987 \end{pmatrix}$
2	21.9112	21.8846	$\begin{pmatrix} 21.3949 & 11.3376 \\ 0 & 7.7181 \end{pmatrix}$	$\begin{pmatrix} 21.3533 & 11.3658 \\ 0 & 7.7190 \end{pmatrix}$	5.1986	5.1983	$\begin{pmatrix} 1.50 \\ 1.80 \end{pmatrix}$	$\begin{pmatrix} 1.5007 \\ 1.8008 \end{pmatrix}$
3	24.7690	24.1672	$\begin{pmatrix} 24.0554 & 11.2340 \\ 0 & 14.9712 \end{pmatrix}$	$\begin{pmatrix} 21.7305 & 12.5178 \\ 0 & 16.0011 \end{pmatrix}$	5.0637	5.0403	$\begin{pmatrix} 0.80 \\ 2.50 \end{pmatrix}$	$\begin{pmatrix} 0.7795 \\ 2.5038 \end{pmatrix}$
4	30.3459	30.1427	$\begin{pmatrix} 12.1598 & 9.2022 \\ 0 & 25.2080 \end{pmatrix}$	$\begin{pmatrix} 12.7446 & 11.3010 \\ 0 & 23.8680 \end{pmatrix}$	5.9312	5.9437	$\begin{pmatrix} 0.75 \\ 1.20 \end{pmatrix}$	$\begin{pmatrix} 0.7444 \\ 1.1974 \end{pmatrix}$
5	36.3180	35.9642	$\begin{pmatrix} 8.5057 & 12.9073 \\ 0 & 24.6596 \end{pmatrix}$	$\begin{pmatrix} 8.1141 & 10.8468 \\ 0 & 25.6805 \end{pmatrix}$	5.4526	5.4477	$\begin{pmatrix} 2.00 \\ 3.00 \end{pmatrix}$	$\begin{pmatrix} 1.9848 \\ 2.9993 \end{pmatrix}$

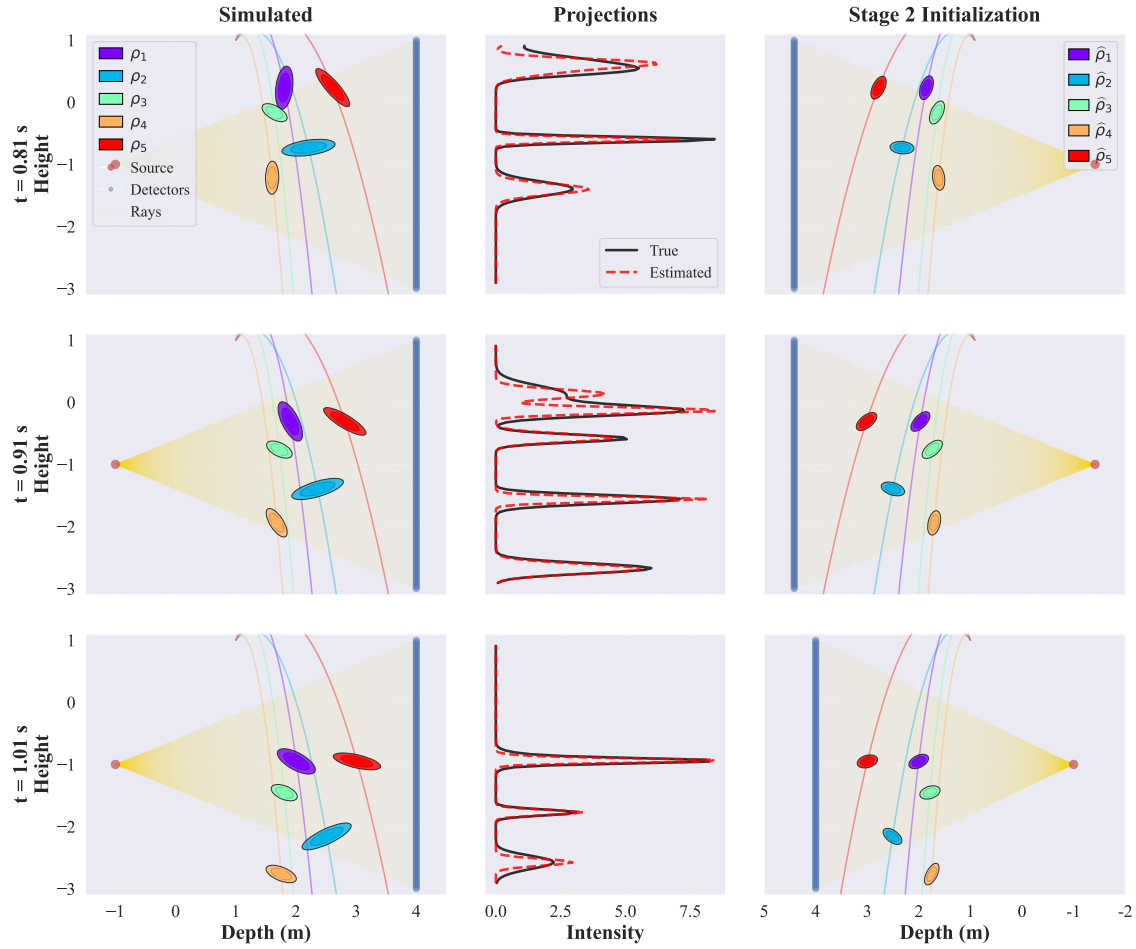


Figure 8.5: Left: The simulated particle states sampled at times $t = 0.81s, 0.91s, 1.01s$. The task is to reconstruct these particles. Right: The Stage 2 initialized estimates plotted as a mirror image. Note the trajectories are exactly those in the right column of Figure 8.4, resulting from the Stage 1 optimization. Center: The combined projections of the two particle sets, by which the discrepancy informs the fit in the particles.

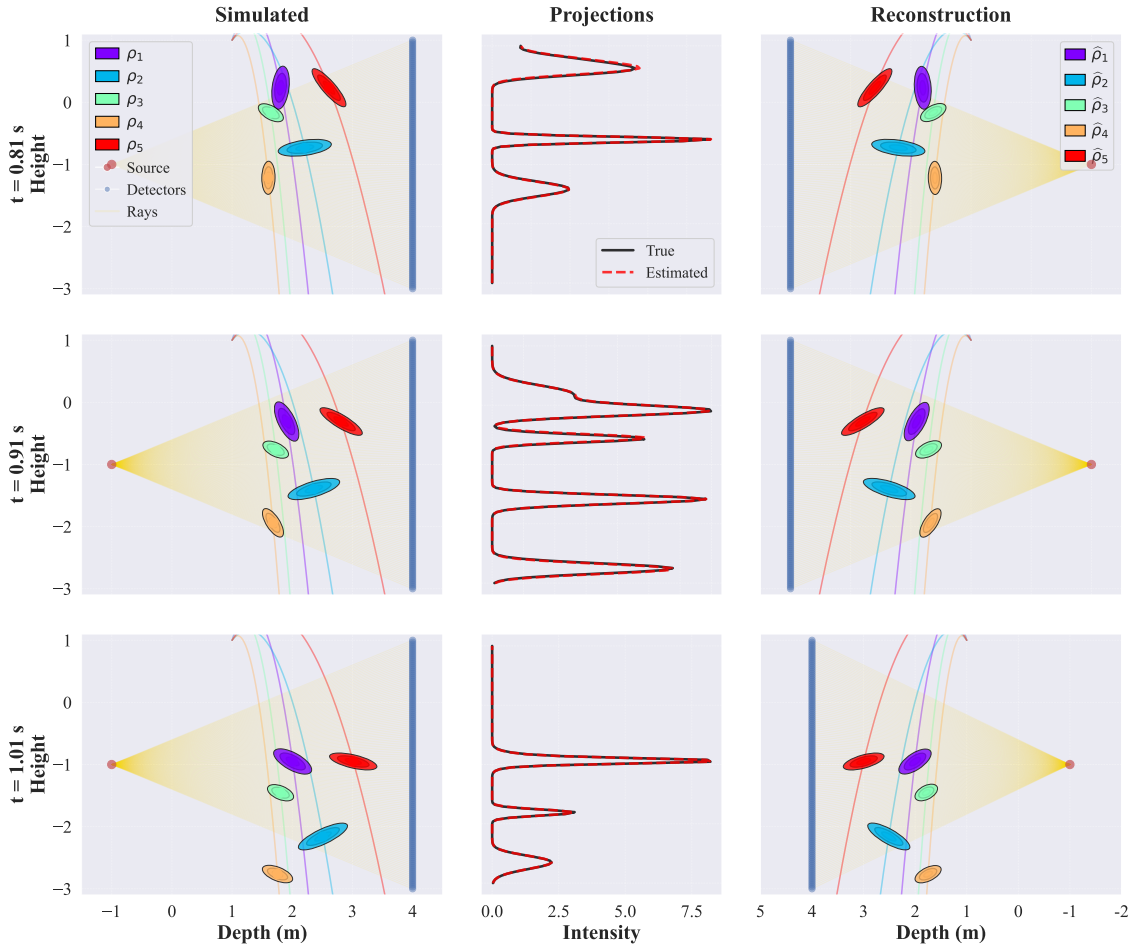


Figure 8.6: The same visualization scheme as in Figure 8.5, except here the right column plots are the output reconstructed particles produced by GMM-CT. The accurate fit in the projections in the center column are generally a necessary but not a sufficient condition for achieving an accurate reconstruction.

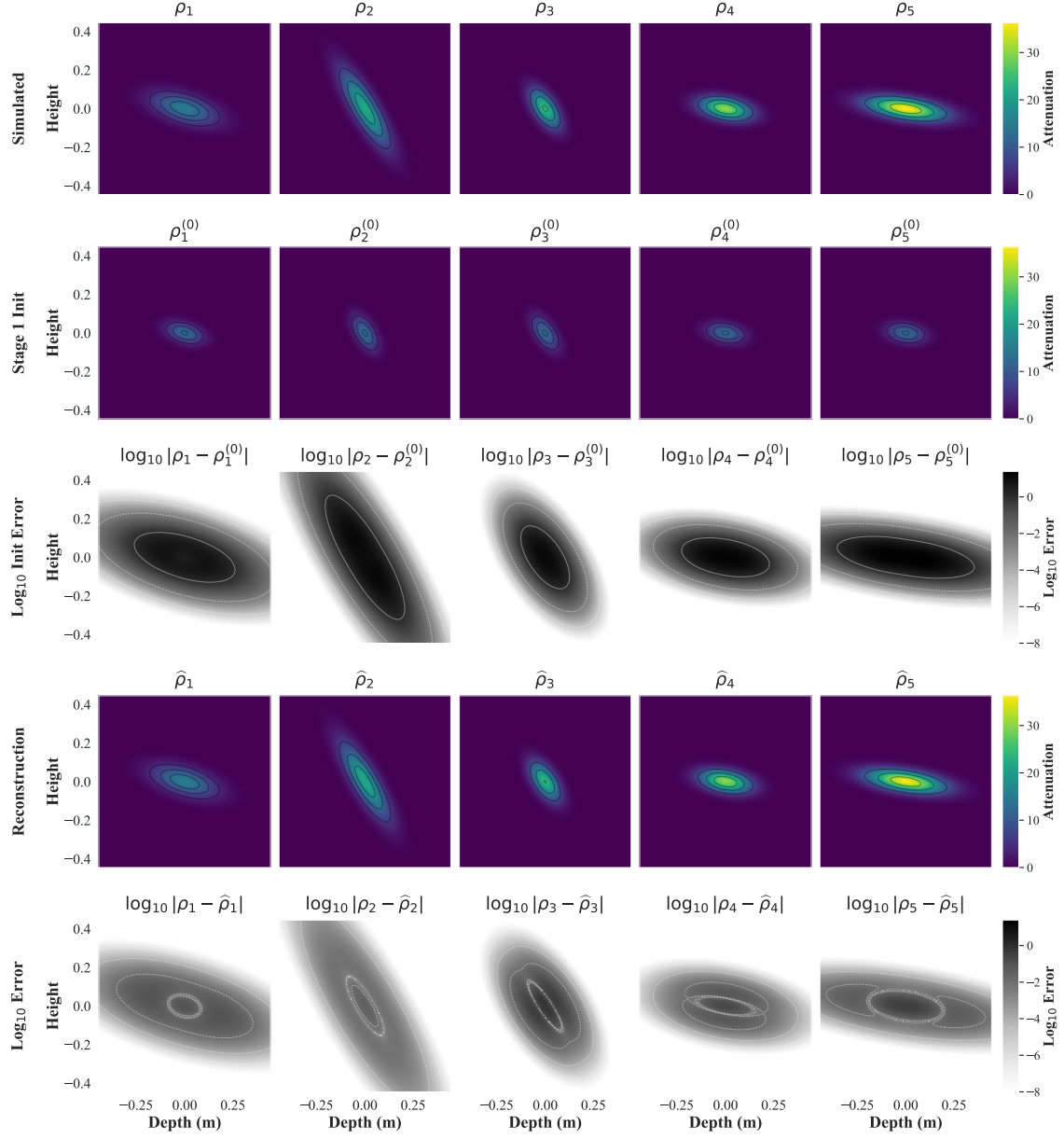


Figure 8.7: First row: the unknown, simulated particles. Second row: the Stage 1 initial-ized estimates (i.e. prior to the Stage 2 initialization or optimization). Third row: The corresponding $\log_{10}$ absolute errors. Fourth row: The reconstructed solutions. Fifth row: The corresponding $\log_{10}$ absolute errors. Throughout, each of the particles are oriented with respect to the rotational motion taken at $t = 0$.

9. Conclusion. By utilizing an informative parametric spatiotemporal model, we have established a theoretical framework and accompanying algorithm (GMM-CT) for automating CT reconstruction. To maximize generality, the model has been designed to operate on linear combinations of a generic class of particles (ρ_n) given by affine transformations of a canonical template function ρ_0 . Due to its closed-form under the ray transform, ρ_0 was chosen as an isotropic Gaussian function, leading to a reconstruction framework based on GMM. GMM are an advantageous choice due to their ability to accurately approximate the mathematical representations of the objects to be reconstructed. One extension of the numerical experiments we have performed is to exploit this feature and attempt to recover more complicated objects in motion than individual Gaussians. In addition to reconstructing alternative objects, the model accommodates customization in motion parameterization and acquisition geometry beyond the examples considered here.

Applicable to Euclidean space of arbitrary dimension, the model propagates each particle with its own independent rotational motion, projectile motion and geometry. The role of the motion is crucial to facilitating the reconstruction process. In particular, the rotation allows different views of the particles to be captured as they traverse the gantry. However, motion alone is not enough to provide a reconstruction. A sufficiently large sampling rate is a necessary prerequisite, and even then, one could get unlucky if the viewing angle and projection times coincide in a way that does not provide a full view. Furthermore, multiple particles have the potential to obscure one another, leading to an identifiability issue and effectively reducing the number of informative projections.

For numerical validation, a 2D, 5-particle GMM case study was simulated and accurately reconstructed. Further informal empirical tests suggest this success is robust across various similar randomly generated GMM configurations. A strength of the way in which we have built GMM-CT is that all of the internal methods are immediately applicable to higher dimensional applications; an exciting next step is the transition to 3D experiments.

A crucial additional complexity that we have not considered in this paper is the presence of noise in the projections. For GMM-CT to be a credible automation tool, its performance in noisy scenarios must be understood. In particular, the algorithm may need to be supplemented with additional methods to successfully detect the modes in the current trajectory optimization procedure. Further extensions include estimating the number of Gaussians N , the reconstruction of *non-rigid* body objects, where the skewness matrix U inherits a time dependency, and also extension ultimately to experiments based on real objects and projections. The framework presented here achieves successful reconstruction for the applications investigated and establishes a formal foundation for these future developments.

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